Physical-Simulation-Based Dynamic Template Matching Method for Remote Sensing Small Object Detection

Yaming Cao, Lei Guo, Fengguang Xiong, Liqun Kuang, and Xie Han

***Abstract*— Object detection is a fundamental challenge encountered in understanding and analyzing remote sensing images. Many current detection methods struggle to identify objects in remote sensing images due to poor feature saliency, diverse directions, and unique viewing angles of small remote sensing targets. As most remote sensing images are captured through overhead photography, the structural contour features of objects in these images remain relatively stable, such as the cross-shaped structure of the aircraft and the rectangular structure of the vehicle. Therefore, we utilize the physical simulation images as prior knowledge to supplement the invariant structural features of the target, extract the target's key geometric features through the dynamic matching method, and fuse it with the features extracted by the neural network to detect the remote sensing small target more effectively in this article. We propose a template matching method based on physical simulation images (TMSI) and add it to the modified Darknet-53 (named TMSI-Net) for small target detection. Based on this, we perform specific transformations on existing templates in terms of target scale and direction to achieve better adaptation to the corresponding goals and propose DTMSI-Net with a dynamic template library (Dynamic TMSI-Net). Experiments on several datasets demonstrate that the DTMSI-Net exhibits higher detection accuracy and more stable performance compared with the state-of-the-art methods, 4% and 2.7% higher than the second-ranked model on the VEDAI and NWPU datasets, respectively.**

***Index Terms*— Dynamic template matching, knowledge template, physical simulation image, remote sensing object detection.**

I. Introduction

With the development of remote sensing technology, high-resolution remote sensing images are becoming increasingly accessible, and their numbers are rapidly increasing [1], [2], [3]. Better and faster processing of remote sensing images is the key to analyzing and understanding remote sensing images. Object detection in remote sensing images has tremendous application value in the field of public security, such as vehicle monitoring and smart city construction [4], [5], [6], [7], [8], [9]. Therefore, the detection of small targets in remote sensing images has gradually become a research hotspot, attracting more and more researchers. The small target mentioned in this article refers to an object that occupies about a dozen pixels in a remote sensing image. However, due to the variety of postures, small size, complexity of background, and special perspective, most of the current detection methods face difficulty in detecting small targets in remote sensing images [10], [11]. In addition to the above factors, weather changes, such as cloud occlusion and light shadows, will also increase the difficulty of detecting small targets in remote sensing images [12].

In recent years, numerous methods for small target detection in remote sensing images have been proposed [13], [14], [15], [16], [17]. Based on the various methods of extracting features from remote sensing images, the approaches for detecting targets in these images can be broadly categorized into two groups: one involves obtaining identified features in the remote sensing image through artificial design and using them for target detection, while the other utilizes convolutional neural networks (CNNs) in deep learning to automatically extract the features in the remote sensing image for target detection [11]. Some researchers have employed handcrafted features to extract characteristics, such as histogram of oriented gradients (HOG) [18], scale-invariant feature transform (SIFT) [19], local binary pattern (LBP) [20], integral channel features (ICFs) [21], and the bag-of-words (BOW) [22]. This method, based on the combination of artificial design features and a classifier, consumes a lot of human resources and a large time cost when designing features. It is difficult to design effective features for small target detection in large-scale remote sensing images.

Deep learning has achieved great success in computer vision, speech processing, natural language processing, and other fields. Among them, the detection model based on deep learning has achieved good results in many datasets, and more and more researchers have applied these models in the field of remote sensing target detection [23], [24], [25]. Modifications based on deep networks to adapt to remote sensing images can fully leverage the powerful ability of the neural networks to automatically extract features. However, there are still some deficiencies in the research on target

Manuscript received 16 October 2023; revised 29 November 2023; accepted 13 December 2023. Date of publication 18 December 2023; date of current version 29 December 2023. This work was supported in part by the Foundation of Shanxi Key Laboratory of Machine Vision and Virtual Reality under Grant 447-110103, in part by the Youth Project of Applied Basic Research Project of Shanxi Province under Grant 202203021222027, and in part by the National Natural Science Foundation of China under Grant 62272426. *(Corresponding author: Yaming Cao.)*

The authors are with the Shanxi Key Laboratory of Machine Vision and Virtual Reality and the Shanxi Province's Vision Information Processing and Intelligent Robot Engineering Research Center, School of Data Science and Technology, North University of China, Taiyuan 030051, China (e-mail: yafei4554@foxmail.com; lei.guo@nuc.edu.cn; hopenxfg@nuc.edu.cn; kuang@nuc.edu.cn; hanxie@nuc.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3344280

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.

detection in remote sensing images based on deep learning. Compared with object detection in natural images, the main difficulty is that most remote sensing images are obtained using overhead photography. Therefore, the remote sensing target may appear to be significantly different from the ground perspective target. In such cases, the effective information and features contained in the target itself are very limited, coupled with complex background, and other reasons, resulting in the limited performance of the method based on deep learning in target detection in remote sensing images. Among these challenges, cloud occlusion and changes in illumination also significantly interfere with remote sensing image target detection. Global cloud data from the International Satellite Cloud Climatology Project Flux Data (ISCCP-FD) show that more than 66% of the Earth's surface area is often covered with clouds [26], [27]. The attenuation and even loss of some image information caused by the cloud not only significantly reduce the quality and utilization of remote sensing data but also pose challenges in the analysis and application of remote sensing images [28], [29].

Considering that the structural contour features of objects in remote images obtained from overhead photography are relatively stable, such as the cross-shaped structure of the aircraft and the rectangular structure of the vehicle [30], this article employs simulation image as prior knowledge to supplement the invariant structural features of the target to overcome the mentioned challenges in remote sensing image target detection. Simulation images of specific angles and directions for the target are obtained through imaging simulation technology, considering the target's 3-D model, material, and atmospheric transmission conditions [31]. The target simulation image obtained by imaging simulation technology contains more target features, especially contour structure features [32]. When combined with deep learning, it can improve the detection results of the model, and adding simulation images can enhance the interpretability of the detection results. We propose a template matching method based on simulated images (TMSI) and use the simulation image as a template to enhance the geometry and position information of the target in the shallow feature map of the neural network to improve the detection performance. The method uses the modified network of Darknet-53 [33] as the backbone network for extracting features. The purpose of modifying the Darknet-53 network structure is to add shallow features for detection while also modifying the network depth to remove deep layers that have no or even negative impact on small object detection. Indeed, we have conducted extensive research on the impact of network structure on tasks related to detecting small targets in remote sensing. The research results indicate that shallow features, including position, shape, and other features, can improve the accuracy of the network for remote sensing small target detection when used in small target detection tasks. All results of the comparative study and the structure of Modified Darknet-53 are presented in the literature [35] In the detection phase, the SSD [34] network structure is referenced, and the multiscale feature map is utilized for upsampling and fusion for detection [35]. The core part of the TMSI-Net is to match the simulation image with the neural network feature map through template matching, thereby enhancing the location and geometry of the target in the feature map. First, according to the type of target to be detected, imaging simulation technology is used to obtain simulation images of different angles and directions under visible. In the process of generating the simulation image, considering the diversity of the target to be detected in the remote sensing image, it focuses on the simulation of the geometric shape of the target. Then, the TMSI algorithm is used to add the obtained simulation image to a specific layer of the network to enhance the geometric information of the target in the feature map. Finally, for dynamic optimization of TMSI-Net using static and fixed template libraries, we propose dynamic TMSI-Net (DTMSI-Net) with dynamic template libraries. It performs specific transformations on existing templates regarding target scale and direction to better adapt to corresponding targets. Compared with the experimental results of advanced methods, the DTMSI algorithm not only performs well in the datasets without cloud occlusion but also maintains a good performance in the presence of cloud occlusion. Our innovations and main contributions can be summarized as follows.

1) In response to the difficulties of poor feature saliency, special perspective, and susceptibility to cloud and mist occlusion in remote sensing target detection tasks, this article conducts a detailed analysis and proposes that strengthening and supplementing target invariant structural features is an effective means to overcome the above difficulties. On this basis, this article innovatively utilizes simulation images based on physical properties as templates with prior knowledge to enhance and supplement the invariant structural features of the target.
2) This article proposes a dynamic template matching method based on physical simulation images (DTMSI-Net) for remote sensing small target detection. As prior knowledge, simulation images can not only enhance the key structural features and spatial information of the target but also supplement corresponding structural features for partially occluded targets. Especially after dynamic adaptive adjustment of simulation images, this method has stronger adaptability. Quantitative, ablation, and qualitative experiments on VEDAI, VEDAI-Cloud, and NWPU have demonstrated the effectiveness of the proposed method.
3) This article also provides a new approach to the organic combination of prior knowledge and deep learning. The powerful feature extraction and expression capabilities of deep learning, combined with prior knowledge and common sense, effectively enhance the accuracy of the network while also improving its interpretability. This provides the possibility for deep learning to be widely used in medical, financial, autonomous driving, and other application scenarios that focus on the interpretability of model results. In addition, this article also contributes to a remote sensing small target detection dataset with cloud and fog occlusion and will make it publicly available for subsequent researchers to conduct corresponding algorithm testing and compare results.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.

The remainder of this article is organized as follows. The related works are described in Section II. In Section III, the proposed methodology for object detection in remote sensing is described. The experimental design and the discussion of the experiment results are introduced in Section IV. Section V presents further discussions and gives a summary of our work.

II. Related Works

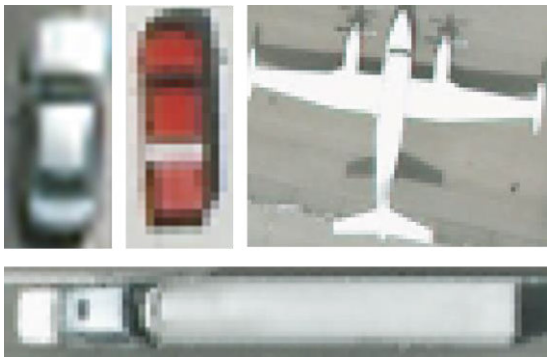
A. Detection of Remote Sensing Objects by Deep Learning

After the great success of object detection methods based on deep learning in the field of computer vision, more and more scholars have used this method to conduct extensive research on the characteristics of objects in remote sensing images, such as insufficient feature saliency, variable viewing angles, and vulnerability to cloud and fog occlusion. Salberg [36] extracted features from a pretrained deep CNN and used them for the automatic detection of seals in aerial remote sensing images. The main reasons why Faster R-CNN [37] does not perform well in the detection of small vehicles in remote sensing images may be due to the coarse feature maps and complex backgrounds [23]. Therefore, they used hyper region proposal network (HRPN) to extract vehicle-like targets with a combination of hierarchical feature maps and replaced the classifier after region proposal network (RPN) [37] with a cascade of boosted classifiers. The negative example mining was also used to reduce false detections. Li et al. [38] believed that prior knowledge about remote sensing images is very useful for the accurate detection of remote sensing targets. Thus, they replaced the convolutional kernels in traditional convolutional networks with selectable large convolutional kernels to increase the receptive field of the network and capture more unique prior knowledge from remote sensing images. In terms of detecting remote sensing small targets with various backgrounds, Pham et al. [39] from the University of South Brittany, France, based on the YOLO V3 model, constructed a lightweight single-stage remote sensing small target detector, YOLO-Fine, by removing the deep feature map of the model and optimizing the network structure. The experimental results show that the shallow layer of CNN contains more features related to the contour, texture, and position of small targets, which is more conducive to the detection of remote sensing small targets. Similarly, to capture more remote sensing target features, Li et al. [40] proposed a general framework of accurate remote sensing object localization constructed with the core idea of maximum image feature exploration. The integrated structure was mainly comprised of cascaded tiny patch correlation (CTPC)-based feature digging, averaging local patch regression (ALPR) centered coarse location acquisition, and instance segmentation-oriented further refinement (ISOFR). This may be the first implementation of the idea of maximum feature exploration in each corner of the object detection architecture. In order to obtain more discriminative features in remote sensing small target detection tasks, Qu et al. [41] proposed a remote sensing small object detection network based on the attention mechanism and multiscale feature fusion and named it AMMFN. The detection head enhancement module (DHEM) and the attention mechanism-based channel cascade (AMCC) module were designed to strengthen the characterization of small object features, reduce the redundant information within the feature layer, and protect small objects from information loss during feature fusion. They also used the normalized Wasserstein distance (NWD) combined with generalized intersection over union (GIoU) as the location regression loss function to improve the optimization weight of the model for small objects and the accuracy of the regression boxes. In response to issues such as the special viewing angle and variable direction of remote sensing targets, Zhang et al. [42] introduced classification and positioning confidence into the sample, preferring to select accurate and consistent anchor points during the model training phase to make the prediction frame fit the target more closely. The introduction of additional parameters has to some extent improved the performance of the model in detecting remote sensing targets with special viewing angles. From the above research work, it can be found that when using deep learning models for remote sensing small target detection, it is necessary to use some network design, optimization methods, or prior knowledge to maximize the effective features of small targets. However, there is a lack of effective analysis of the differences between different features of the target. The features extracted by the neural network are arranged in the form of a 2-D grid, which makes the features extracted by the neural network have good translation invariance, but they perform poorly in terms of rotation invariance and scale invariance. During the translation, rotation, and scaling processes, the structural features of the target will be well preserved, but the neural network model is prone to losing the structural features of the target when extracting features during the above processes.

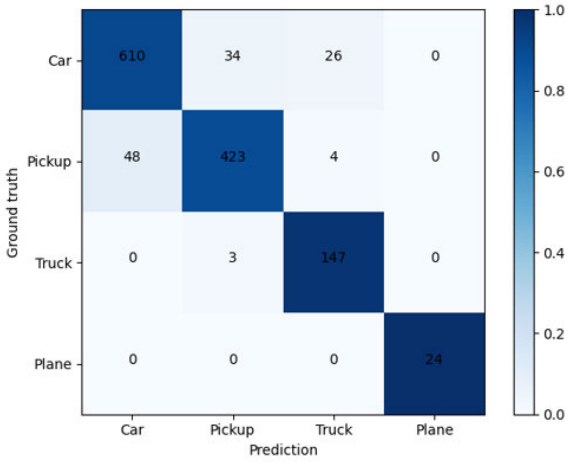
B. Structure-Invariant Features in Remote Sensing Objects

For the detection of small targets in remote sensing images, the most critical features are the geometric features and spatial structure of the target. The geometric features and spatial structure of small targets are not only the key information to distinguish different types of targets but also for achieving effective detection of similar targets [43], [44]. Different objects in the remote sensing image as shown in Fig. 1 exhibit the greatest similarity in structural features. These apparent structural features can be determined as a strong basis for distinguishing the remote sensing objects. The effective extraction and utilization of the structural features of remote sensing objects are of great help in downstream tasks, such as precise localization and fine-grained classification [45]. Han et al. [46] used the region-of-interest (RoI) part to extract the structural feature and proposed rotation equivariant features to describe the direction of the target. When the target in the input image undergoes a certain angle rotation, some features will also undergo corresponding changes, which are the rotation equivariant features of the target. Zhang et al. [47] integrated global context information into target detection and designed a spatial-and-scale-aware attention module that guides the network to focus on more informative regions and features as well as more appropriate feature scales. In addition

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.



(a)



(b)

Fig. 1. We used nearly half of the "Car," "Pickup," "Truck," and "Plane" instances in the VEDAI dataset as input data for the preliminary experiment to confirm the validity of geometric structure features for accurately detecting the same type of remote sensing target, as shown in Fig. 1(a). In the experiment, the geometric contour features of input data were extracted and classified based on the template matching method using simulation images. The experimental results in the confusion matrix in Fig. 1(b), and the sum of each row of numbers in the confusion matrix is the total number of such targets in the experiment. The results indicate that although there are differences in color, direction, and size between instances of the same type of target, they have strong consistency in geometric contour features.

to extracting the invariant structural features of the object through the method of model structure design, many works use rotating quadrilaterals to represent the object boundary, such as using angle offset and key point heat map, to establish the angle mapping of the rotating object. Considering that the horizontal rectangular box has a poor representation of tilted targets in multiple directions and the discontinuous rotation angle makes it difficult to represent objects using a rotating rectangle, Xu et al. [48] located a quadrilateral to represent an object by learning the offsets of four points on a non-rotating rectangle. The experimental results show that the quadrilateral obtained by learning four points and tilt angles can better represent the target boundary. Zhu et al. [49] combined deformable convolution to propose a rotating object representation method based on self-organizing point sets, effectively solving the semisupervised problem of existing methods in dense and inclined remote sensing targets in any direction, and improving the model's ability to accurately locate. The above method focuses on the design of prediction boxes for tilted targets in any direction, which is beneficial for improving the intersection and union ratio of detection results. To better utilize structural invariant features for remote sensing object detection, Zhu et al. [30] proposed a graph-focusing aggregation network based on the idea of graph convolution to represent the structural features. Their experiment results showed that graph structure had significant advantages over Euclidean feature space under CNN in expressing such structural features. By using deformable convolutions or graph convolutions instead of traditional convolutions in the model, more structurally invariant features of the object can be extracted. However, for some remote sensing images where the target itself lacks invariant structural features due to cloud interference or other reasons, the above methods perform poorly due to a lack of prior knowledge about the target structural features. Therefore, we attempt to use simulated images as knowledge templates for remote sensing small target detection, hoping to not only enhance the invariant structural features present in remote sensing images but also supplement the aforementioned features lost by the target due to occlusion.

C. Simulation Image in Deep Learning

The method of using simulation images and deep learning to improve model performance is becoming more and more applied. Tian et al. [50] took the specifically designed virtual datasets as resources to train object detectors and as a tool to produce feedback on the performance of trained models during the testing phase. Zhang et al. [51] proposed a novel sensor-related image synthesis framework to address the lack of on-orbit remote sensing images, dubbed the remote sensing-image synthesis pipeline (RS-ISP). They used a novel pipeline within RS-ISP to model the physical process of noise generation during image capture using specific sensors and designed a detection-oriented image coordination model to ensure distribution consistency between the generated image and the real image. Cao et al. [32] used a simulation image as a template to obtain geometric invariance-based related features of the input image through template matching and integrate them with the high-level features extracted by the neural network, making the training of the model faster, with fewer oscillations, and greatly improving the model's resistance to adversarial examples. Dai et al. [52] used the simulation image to expand the training set and completed the transfer learning training of the model. Simulation technology can be used to obtain simulation images under different views, angles, and weather conditions of the target. Compared to other virtual images, simulation images have clear physical models and physical parameters, and the generated angles and directions are diverse and controllable. Therefore, it is considered to overcome the difficulties in remote sensing image target detection by applying simulation images to deep learning.

III. Methodology

A. Simulation Image

The simulation image used in the TMSI algorithm is shown in Fig. 2(b). These simulation images are generated under the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.

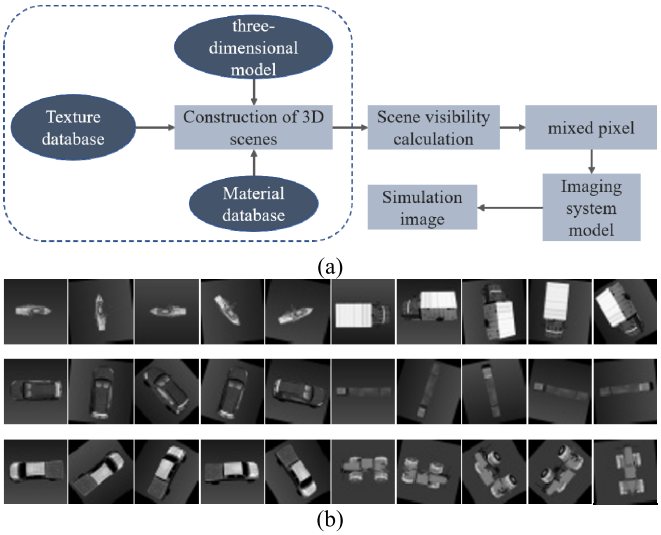


Fig. 2. (a) Process of the imaging simulation framework. (b) Examples of simulation images obtained by the imaging simulation framework.

visible light waveband using the imaging simulation framework in Fig. 2(a), and the size and color of the simulation images are adaptively modified. Considering that the target background in the remote sensing image is more complex and changeable, for example, many boats in the VEDAI dataset [53] are on land instead of water as the background that is usually recognized. The main body of the simulation image is the target, and no specific background is added. Then, to eliminate the interference of the target color diversity to the matching as much as possible, and at the same time, considering that the simulation image is matched with the feature image instead of the original image as input, all the simulation images after adjusting the size are converted to grayscale images. In the process of simulation, by adjusting the horizontal angle and the overhead angle of the target 3-D model, a multiangle and multiscale target simulation image is obtained. There are nine different classes in the VEDAI dataset: plane, boat, car, truck, tractor, camping car, van, pickup, and the other category. To compare the impact of the presence or absence of a simulation image as a template on the detection accuracy, and to save the computational overhead during matching, not every class has a corresponding simulation image. We simulated six of them, as shown in Fig. 2 from left to right, which are cars, pickups, tractors, trucks, vans, and boats. Finally, 24 simulation images were obtained as templates to match the feature maps.

B. TMSI-Net

Template matching is a basic and original method in digital image processing [54]. The method of template matching can measure the geometric invariance between the image and the template. Specifically, in addition to subtracting their means, they are also divided by their respective variances. The operation on the template $T(i, j)$ and image $I(i, j)$ can be seen as (1) and (2), where (w, h) and (m, n) are the width and height of the template and image, respectively. The image and the template are standardized after the two steps. From the expression of the correlation coefficient $R(i, j)$ as (3), we can get the most similar area between the target image and the template and the corresponding correlation coefficient

$$T'(i, j) = T(i, j) - \frac{1}{w \times h} \sum_{i', j'} T(i', j') \tag{1}$$

$$I'(i, j) = I(i, j) - \frac{1}{m \times n} \sum_{i', j'} I(i', j') \tag{2}$$

$$R(i, j) = \frac{\sum_{i', j'} (T'(i, j) \cdot I'(i + i', j + j'))}{\sqrt{\sum_{i', j'} T'(i, j)^2 \sum_{i', j'} I'(i + i', j + j')^2}}. \tag{3}$$

The essence of the TMSI algorithm is to use the simulation image as a template to enhance the structurally invariant features of small targets in the feature map. The input of the TMSI algorithm is the feature map generated by CNN and the simulation image as a template. Templates of different classes and angles are used to enhance different targets. To avoid nonkey features such as color diversity within the class from affecting the matching effect of the template and considering that the feature map after downsampling and big stride convolution will lose too many small target features, the feature map before the first pooling layer is used as the input of the TMSI algorithm. In the training process, we set the batch size to 8, and the first convolutional layer has 32 convolution kernels. Thus, the shape of the feature map F is $8 \times 32 \times 512 \times 512$. The shape of the template library composed of simulation image S is $24 \times 1 \times 30 \times 30$. In the process of template matching, the hyperparameter δ is used to measure the correlation coefficient r. If the r in a certain area is greater than δ, the area is considered to be the most likely category corresponding to the template and, then, the parameter directional enhancement of the area. In the experiment, we set δ to 0.6. The structure of the TMSI can be seen in Fig. 3. When the TMSI layer in the upper right of Fig. 3 is connected to the Modified Darknet-53 on the left, TMSI-Net can be obtained. The whole structure is similar to the residual structure. The residual structure is finally vector splicing, and the output of TMSI is the result of the addition of the input and the corresponding area. After the feature map is matched with all templates, the areas more likely to have small targets are strengthened, and more key features and information about small targets are used for detection

$$\text{loss} = \text{loss_}xy + \text{loss_}wh + \text{loss_conf} + \text{loss_class}. \tag{4}$$

After obtaining the enhanced feature map F from the simulation images, the model continues to perform forward propagation. The loss function of TMSI-Net is shown in (4). loss_xy and loss_wh represent the coordinate error and size error of the object, respectively, and are obtained by calculating the root mean square of the difference between the predicted value and the true value. loss_class and loss_conf represent the loss of target category and confidence. In the calculation process loss_class, the label smoothing function (label-smooth) is used for category balancing. In the training process, a large number of simple background samples occupy the main part of the loss function, so in the calculation process

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.

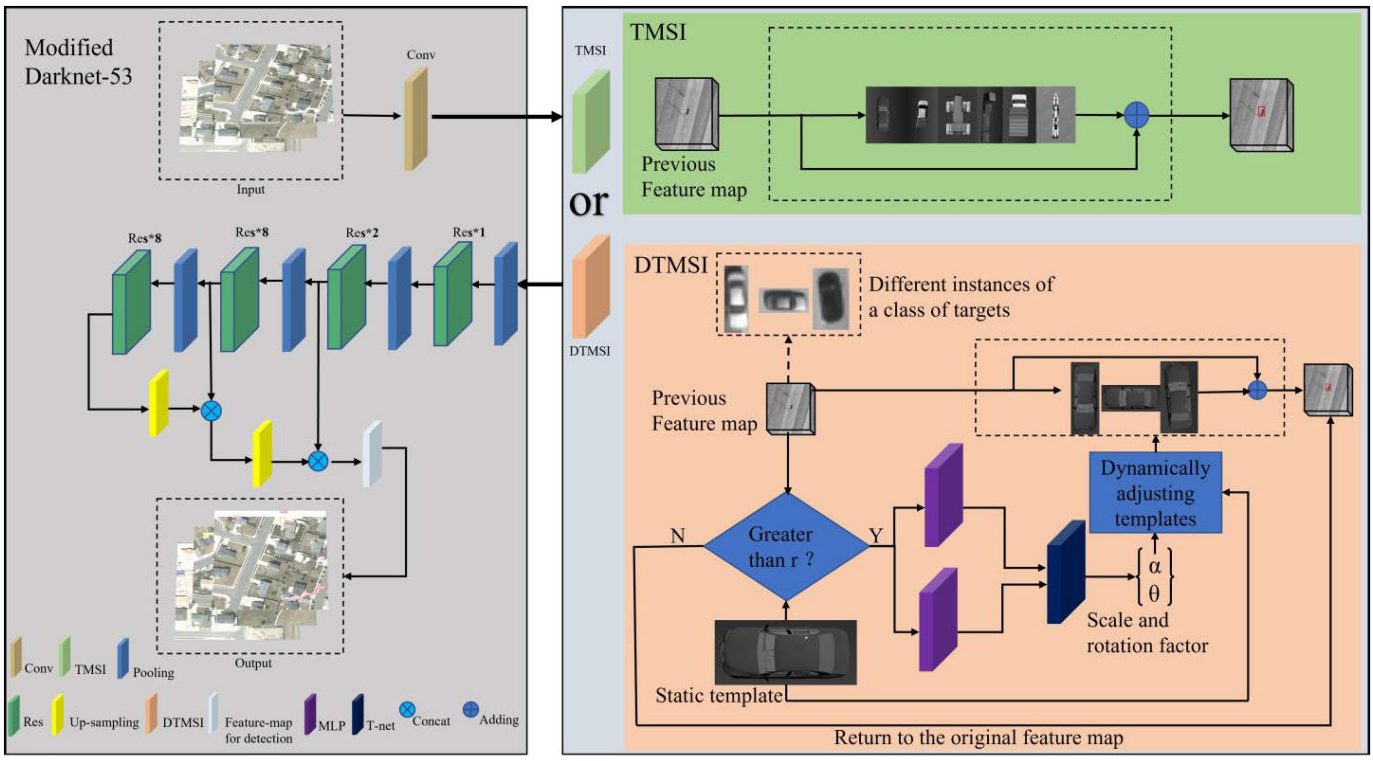


Fig. 3. Structure of Modified Darknet-53, and schematic of TMSI and DTMSI. TMSI and DTMSI are easily pluggable components in the network. When TMSI is plugged into the Modified Darknet-53, the model is TMSI-Net. Similarly, when the component in the Modified Darknet-53 uses DTMSI, the model is DTMSI-Net. TMSI and DTMSI layers do not access the network simultaneously, and they only have one access network to obtain the corresponding model (TMSI-Net or DTMSI-Net).

loss_conf, the focal loss is used to balance the number of positive and negative samples.

C. DTMSI-Net

When using the simulation framework to generate the simulation image of the target, it is usually necessary to consider the material, texture spectrum, and 3-D information of the target. If the above simulation parameters can be accurately identified for the target, a simulation image that is highly similar to the target can be obtained, and the simulation image will contain rich features and information about the target. For some tasks that focus on the target texture material or require a high degree of precision of the target, the accurate identification of all simulation parameters and dynamic real-time simulation has important research significance. However, for the detection task of small remote sensing targets, it is very difficult to accurately obtain the material and texture information of all targets according to the changes of the remote sensing images to be detected. Moreover, for the same type of target, there are large differences in materials, textures, color, and so on, so this information cannot be used as the main basis for detecting this type of target. The remote sensing images mentioned in this article are all obtained in the visible spectrum, so the spectral information cannot be the effective information and features that the simulation technology can provide for the detection of small targets. In remote sensing images, the geometric features and spatial structure of the same type of targets can help distinguish the key features and information of such targets. Considering that remote sensing images are usually from a top-down perspective, the 3-D model of the target is obtained from a top-down perspective. The simulated images can provide help for remote sensing small target detection tasks in terms of geometric features and spatial information.

Considering that the limited static template will indeed limit the feature extraction of feature maps, we continue to optimize TMSI-Net by dynamizing the simulation image as a knowledge template and performing specific transformations on the existing template from the perspective of the target scale and direction, which has achieved the purpose of better adapting to the corresponding target. This proposes a DTMSI-Net with a dynamic template library. The schematic of dynamically adjusting templates is in Fig. 3. When the DTMSI layer at the bottom right in Fig. 3 is connected to the Modified Darknet-53 on the left, DTMSI-Net can be obtained.

The specific idea of constructing a dynamic simulation image template library is to use the general materials and textures of this type in the visible spectrum according to the existing 3-D model of the type of target and obtain a simulation image from the perspective of the camera as a static template. First, the static template is used to perform preliminary matching detection on the target. When the correlation between the static template and a certain area of the feature map reaches a certain threshold, the template is rotated and scaled accordingly. The template is dynamically adjusted by the scaling factor α and the rotation angle θ, and they are optimized based on the direction with the greatest correlation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 TMSI

Input: The feature maps F to be matched, the shape of it is $bs \times ch \times w \times h$, where bs is the batch size, ch is the number of channels F, $w \times h$ is the shape of each channel, a threshold δ, where $0 < \delta < 1$, the grayscale simulation images as template library S, the shape of it is $N \times 1 \times n \times m$, where N is the number of simulation images, $n \times m$ is the shape of each simulation image.

```
get N (number of S)
for i = 1, 2, ..., bs, do
  for j = 1, 2, ..., ch do
    I = F(i, j, :, :)
    for t = 1, 2, ..., N do
      T = S(t, 1, :, :)
      standardize T and I use formulas (1) and (2)
      for (x, y) = (0, 0), (0, 1), ..., (w − n, h − m)
       do
          use formula (3) to calculate the correlation
          coefficient r between I and T, and get the
          input image I coordinates (x_r, y_r)
          corresponding to the current r
          if r > δ:
            I(x_r : x_r + n, y_r : y_r + m) += r × I(x_r : x_r + n, y_r : y_r + m)
  end for
```

Output F

between the template's geometric contour and the target to be detected. Finally, different templates corresponding to different morphological targets in the feature map are obtained, which can effectively improve the detection results, as shown in Fig. 3.

In Fig. 3, the MLP module is a multilayer perceptron with one hidden layer and 64 neurons. Two MLP modules are used to preprocess the dataset and extract the features for the feature map area and basic template corresponding to the threshold. The processed data and features are aligned through a feature alignment network T-net. T-net is a multilayer perceptron with two hidden layers, 64 and 32 neurons. When optimizing MLP and T-net in DTMSI-Net, the variation $(1\text{-}R(i, j))$ of the correlation coefficient $R(i, j)$ in (3) between the template and the feature map is used as the loss function. When calculating $R(i, j)$, T is obtained by dynamically adjusting the regular template T_0 through α and θ. When the template rotates θ counterclockwise with the center as the origin, the corresponding rotation matrix is β. The corresponding relationship between θ and β is as follows:

$$\beta = \begin{Bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{Bmatrix}. \tag{5}$$

Thus, the dynamic template T can be obtained through the transformation of T_0 in (6)

$$T = \alpha T_0 * \beta. \tag{6}$$

The loss function during the dynamic adjustment of the template is shown in (7). As a part of the loss of the whole detection network, this loss function, together with the location loss and category loss of the target in formula (4), constitutes the whole loss function of the network. The SGD method is used to optimize the MLP and T-net parameters by using the loss function, to learn the scaling factor α and rotation angle θ of the dynamically adjusted template

$$\text{loss} = 1 - \frac{\sum_{i',j'} ((\alpha T_0 * \beta)(i, j) \cdot I(i + i', j + j'))}{\sqrt{\sum_{i',j'} (\alpha T_0 * \beta)(i, j)^2 \sum_{i',j'} I(i + i', j + j')^2}}. \tag{7}$$

TABLE I
STATISTICS OF THE VEDAI DATASET

Class name	Total	Orientation
Boat	170	[-π π]
Camping car	390	[0 π]
Car	1340	[-π π]
Other	200	[0 π]
Pickup	950	[-π π]
Plane	47	[-π π]
Tractor	190	[-π π]
Truck	300	[-π π]
Vans	100	[-π π]

IV. EXPERIMENTS

A. Datasets and Settings

We perform experiments over three datasets, VEDAI [53], VEDAI-Cloud, and NWPU [55]. Vehicle detection in aerial imagery (VEDAI) is a dataset of vehicle detection in remote sensing images, and examples of VEDAI are shown in Fig. 4(a). The dataset is formed by dividing the satellite image in a large field of view into 1024 × 1024 pixels and all images have been taken from the same distance to the ground. An equal number of 512 × 512 pixels images are obtained after downsampling each of the above images to make the targets smaller. As a dataset for benchmarking the object detection algorithm in an unrestricted environment, the dataset contains different variability in addition to very small vehicles, such as multiple directions, light and shadow changes, mirror reflection, or occlusion. To study the effect of the improved model on the detection of small targets in remote sensing images, this article only uses 512 × 512 pixel images for experiments and does not resize and crop the images. The VEDAI dataset contains nine different classes of vehicles, namely, the "plane," "boat," "car," "truck," "tractor," "camping car," "van," "pickup," and the "other" category. There is an average of 5.5 vehicles per image, and they occupy about 0.7% of the total pixels of the images. The target size in VEDAI is a typical remote sensing small target, whether defined in terms of relative or absolute scale. The number of various types of targets in the dataset is shown in Table I. The entire dataset is divided into a training dataset and a validation dataset at a ratio of 5:1 (training to validation, respectively), randomly selected from the VEDAI dataset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 4. (a) VEDAI examples. (b) VEDAI-Cloud examples.

As introduced in Section I, global cloud data from the ISCCP-FD show that more than 66% of the Earth's surface area is often covered with clouds. Compared with natural images, remote sensing images are more susceptible to cloud cover and changes in lighting. Thus, accurately detecting small targets in remote sensing images under cloud and fog interference has become a challenge that must be faced and solved [28], [29]. However, to our knowledge, most of the publicly available remote sensing object detection datasets are collected under good visibility and without cloud or mist interference, lacking a dataset for remote sensing object detection under cloud or fog interference conditions. This article considers constructing a new dataset VEDAI-Cloud based on the VEDAI dataset by artificially increasing cloud and fog interference to better approach real remote sensing target scenes and test the performance of the model in real remote sensing target scenes. To make the effect as close as possible to the effect in a real scene after being obscured by clouds and mist, the clouds and mist in the dataset are rendered using a cloud and mist filter in Photoshop. The transparency of the filter corresponds to the thickness of the cloud, and the transparency of the filter varies randomly between 0% and 70%. By adding this filter to all images in the VEDAI dataset, the VEDAI-Cloud dataset is obtained, such as in Fig. 4(b). We will soon make the VEDAI-Cloud dataset publicly available for other researchers to verify methods and compare results on this dataset.

To comprehensively verify the effectiveness of the method proposed in this article for remote sensing object detection, we also conducted experiments on YOLO V3 [33], SEN [56], YOLO-Fine [39], DEAN [57], YOLO V8n [58], YOLO V8x [58], SuperYOLO [59], MFE-Framework [40], LSK-Net [38], TMSI-Net, and DTMSI-Net on the NWPU dataset. The NWPU dataset contains 800 high-resolution satellite images cropped from the Google Earth and Vaihingen datasets, manually annotated by experts. NWPU is a challenging dataset for detecting ten types of remote sensing objects, which can be used for single and multitype remote sensing object detection.

The definition of IoU is shown in (8) and a detected bounding box B and a ground truth G is considered as a correct match if

$$\text{IoU} = \frac{\text{Area}(B \cap G)}{\text{Area}(B \cup G)} \geq 0.5. \tag{8}$$

True positive (TP) means a predicted object whose IoU with the matched ground truth (GT) is larger than the threshold. False positive (FP) is just the opposite. After matching all the predicted objects, the remaining unmatched GT is called a false negative (FN). The precision and recall are calculated by the following equation:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{9}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{10}$$

Average precision (AP) and mean AP (mAP) are adopted to evaluate the performance of the models on the remote sensing object detection task. AP can be derived by calculating the area between the precision–recall (PR) curve and its coordinate axis such as (11). AP50 in the later text means that the IoU threshold is greater than 0.5, as shown in (8). mAP is needed for a multicategories task such as (12), where N refers to the number of object categories

$$\text{AP} = \int_0^1 P(R)d(R) \tag{11}$$

$$\text{mAP} = \frac{1}{N}\sum_{i=1}^{N}\int_0^1 P_i(R_i)dR_i. \tag{12}$$

The computer specifications for the experiment are as follows: Ubuntu 16.04 system, the deep learning framework used is Pytorch, and the GPU used is NVIDIA GeForce TITAN V. Unlike MLP and T-net using the SGD method as the optimizer, Adam is used as the training optimizer in the TMSI-Net and DTMSI-Net, where the coefficients β_1 and β_2 use the default values of 0.9 and 0.999, respectively. The training of all models in this article is set to 200 epochs and the batch size is set to 8. If the change in mAP (IoU $\geq$ 0.5) of the model is less than 0.1% over 20 consecutive rounds of training, training will be terminated early. The model weight with the highest mAP in all rounds is used as the final training result for the model.

B. Quantitative Experimental Results and Ablation Studies

We compared the quantitative experimental results of all models using the VEDAI dataset. All models were trained to the optimal states on the VEDAI dataset, with inference test results shown in Table II. By comparing the results with other models in Table II, we can see that the DTMSI-Net proposed in this article performs well in small target detection tasks in remote sensing images. DTMSI-Net performed well, approaching the highest accuracy of most categories in other models, and achieved the highest accuracy on mAP, precision, and recall, which are 4%, 5%, and 5.7% higher than those of the second-ranked model, respectively. In terms of inference speed, SEN with the simplest network structure performs

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.

TABLE II
DETECTION RESULTS (AP50/% AND mAP/%) OF ALL MODELS ON THE VEDAI DATASET

Models	Car	Truck	Tractor	Camping Car	Other	Van	Boat	Plane	Pickup	FPS	mAP @.5	P	R
YOLO V3	69.1	32.8	52.5	54.7	31.4	21.1	4.9	93.2	44.7	19.2	44.9	46.7	41.5
SEN	60.0	24.5	37.0	54.8	19.0	56.6	26.5	**100.0**	52.0	**35.4**	47.8	49.1	46.3
YOLO-Fine	83.5	33.5	39.0	62.7	47.9	49.6	18.2	93.3	70.4	22.1	55.3	57.2	52.9
DEAN	85.5	49.6	58.8	72.9	**57.5**	**81.3**	46.9	85.1	70.2	28.6	67.5	69.3	65.1
YOLO V8n	83.7	49.6	56.4	65.2	43.3	34.4	30.3	99.5	66.3	29.7	58.7	61.4	55.8
YOLO V8x	80.3	58.6	63.9	66.1	36.9	22.2	42.6	99.5	64.7	23.1	59.4	62.3	57.6
SuperYOLO	90.3	**68.6**	79.5	76.7	53.9	70.3	**58.1**	-	**82.7**	-	72.5	73.5	69.9
MFE-Framework	82.1	29.6	47.8	63.6	28.9	46.0	18.2	**100.0**	76.8	23.9	54.8	56.2	52.0
LSK-Net	83.0	53.9	79.4	**85.0**	45.6	57.4	55.6	**100.0**	77.5	21.8	70.8	74.5	66.1
TMSI-Net(ours)	93.9	46.4	84.0	75.4	43.1	51.7	43.4	**100.0**	69.1	20.2	67.4	68.7	65.9
DTMSI-Net(ours)	**97.1**	66.3	**89.7**	84.2	55.8	62.9	54.7	**100.0**	77.8	21.5	**76.5**	**79.5**	**75.6**

the fastest detection; however, its detection accuracy is poor. It takes 2 h and 15 min for the TMSI-Net model to converge after 200 rounds of training. However, due to the inclusion of new MLP and T-net dynamic template adjustment modules in the training process of DTMSI-Net, it takes 2 h and 35 min for DTMSI-Net to reach convergence. During the inference stage, DTMSI-Net has a faster detection speed compared to TMSI-Net. This is mainly because during the inference stage, DTMSI-Net does not need to optimize the parameters of the added dynamic template modules, and there are fewer templates used to enhance the target invariant structural features. A DTMSI-Net method still has significant optimization space in terms of detection speed.

The experimental results of TMSI-Net and DTMSI-Net in Table II indicate that adding simulation images as knowledge templates to the network can some extent supplement the geometric and spatial distribution invariant structural features of remote sensing small targets. When the simulation image is optimized to the adaptive stage for dynamic template matching, it can provide more small target key invariant structural features and achieve the best detection performance of models.

Considering that not all categories have corresponding simulation images as templates, we conducted further statistical analysis to explore the changes in accuracy between categories with and without simulation images as templates, using the detection results of Modified Darknet-53, TMSI-Net, and DTMSI-Net, as shown in Table III. $\triangle \text{mAP}_{\text{sim}}$ in Table III represents the average increase in the accuracy of categories after adding simulation images as templates, while $\triangle \text{mAP}_{\text{no-sim}}$ represents the average increase in the accuracy of categories without simulation images as templates. From the comparison of detection results between TMSI-net and Modified Daknet-53 in various categories, it can be found that after adding static simulation images as templates, the accuracy of all categories except for "Van" and "Pickup" has improved. The reason for the poor accuracy of "Van" and "Pickup" in TMSI-Net may be that there are significant differences in size and direction between the target in this category and the corresponding template. At the same time, the structural contours of "Van" and "Pickup" category targets are more easily confused and difficult to distinguish compared to "Truck" and "Tractor." From the detection results of various categories in DTMSI-Net, it can be seen that after the simulation image as a template is dynamically transformed, the accuracy of all categories has been improved. This also suggests that the adaptive template can not only supplement the structural contour features of the target but also better capture the size and direction features of the target. From the comparison of $\triangle \text{mAP}_{\text{sim}}$ and $\triangle \text{mAP}_{\text{no-sim}}$ in TMSI-Net and DTMSI-Net, it can be observed that overall, the accuracy improvement of categories that incorporate simulation images as knowledge templates is better than those without templates, with an increase of 5.9% and 8.6%, respectively.

TABLE III
COMPARISON OF RESULTS (AP50/%) BETWEEN CATEGORIES WITH SIMULATION IMAGES AS TEMPLATES AND CATEGORIES WITHOUT TEMPLATES FROM MODIFIED DARKNET-53, TMSI-NET, AND DTMSI-NET

	Modified Darknet-53	TMSI-Net	DTMSI-Net
Car	83.6	93.9(+10.3)	97.1(+13.5)
Truck	22.9	46.4(+23.5)	66.3(+43.4)
Tractor	55.3	84.0(+28.7)	89.7(+33.6)
Van	58.7	51.7(-7.0)	62.9(+4.2)
Boat	34.2	43.4(+9.2)	54.7(+20.5)
Pickup	77.3	69.1(-8.2)	77.8(+0.5)
$\triangle \text{mAP}_{\text{sim}}$	baseline	**+9.4**	**+19.3**
Camping Car	63.9	75.4(+11.5)	84.2(+20.3)
Other	44.0	43.1(-0.9)	55.8(+11.8)
Plane	100.0	100.0(+0)	100.0(+0)
$\triangle \text{mAP}_{\text{no-sim}}$	baseline	+3.5	+10.7

To comprehensively assess the efficacy of dynamic template matching methods based on simulated images for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.

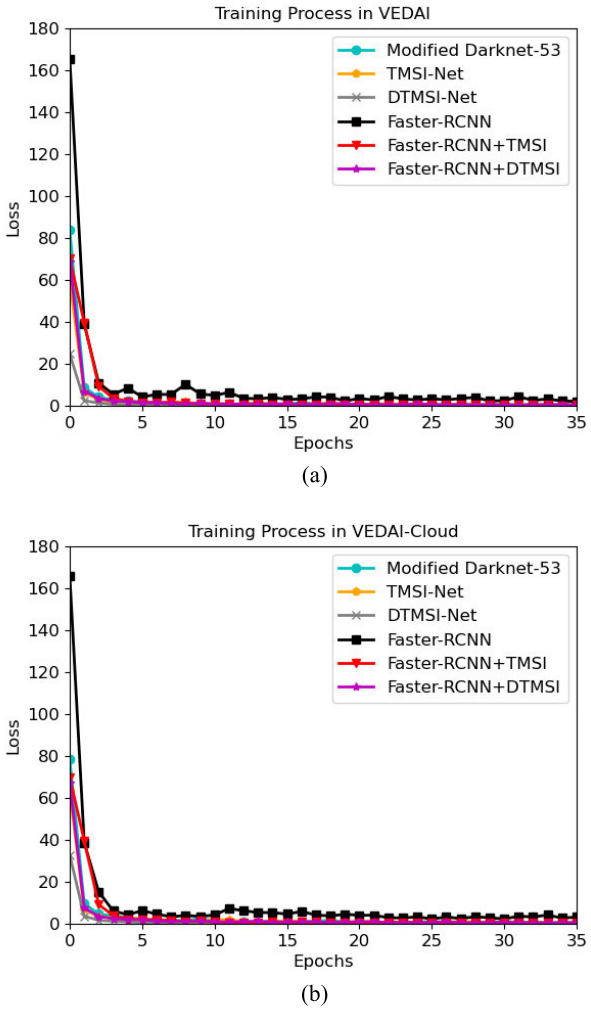


Fig. 5. (a) Loss changes of all models in the ablation experiment trained in the VEDAI dataset (the first 35 epochs). (b) Loss changes of all models in the ablation experiment trained in the VEDAI-Cloud dataset (the first 35 epochs).

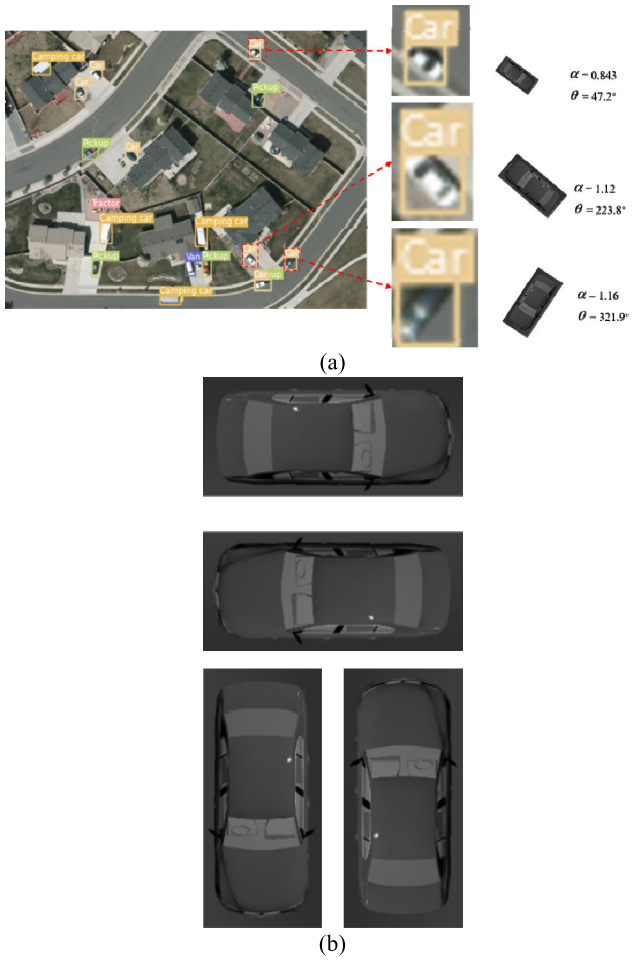


Fig. 6. (a) Template dynamic adjustment diagram. (b) Limited templates in TMSI-Net.

remote sensing small target detection, YOLO V3, SEN, YOLO-Fine, DEAN, YOLO V8n, YOLO V8x, SuperYOLO, MFE-Framework, LSK-Net, TMSI-Net, and DTMSI-Net models were evaluated using VEDAI-Cloud and NWPU datasets. All models were trained to their optimal state on these datasets, and the inference test results are presented in Table IV. The mAP@.5 from the table refers to the mAP of all classes at IoU = 0.5, while mAP@0.5:0.95 represents the average mAP for IoU values ranging from 0.5 to 0.95. From the comparison of results, it can be observed that the DTMSI-Net proposed in this article achieved the highest map on both datasets, surpassing LSK-Net by 9.4%, 5.2%, 2.7%, and 2.8%.

By comparing the experimental results of the VEDAI and VEDAI-Cloud datasets in Tables III and IV, it can be found that cloud and mist occlusion has a significant interference on remote sensing small target detection, greatly reducing the detection accuracy of YOLO V3, SEN, YOLO-Fine, DEAN, and SuperYOLO algorithms. This is mainly because the target itself in the image lacks invariant structural features due to cloud and fog interference. In contrast, the TMSI-Net and DTMSI-Net proposed in this article, which uses simulation images as knowledge templates, exhibit good stability in small target detection tasks in remote sensing images. This also suggests that applying simulation images as knowledge templates to deep learning networks can provide more key invariant structural features of the target; in particular, the adaptively adjusted dynamic template matching method based on simulation images can more targeted improve and supplement the geometric features and spatial structures of the target.

TABLE IV

COMPARISON RESULTS (mAP/%) WITH STATE-OF-THE-ART METHODS ON VEDAI-CLOUD AND NWPU DATASETS

Models	VEDAI-Cloud		NWPU	
	mAP@.5	mAP@.5:.95	mAP@.5	mAP@.5:.95
YOLO V3	30.6	8.4	68.7	42.5
SEN	40.0	15.7	60.9	38.4
YOLO-Fine	52.9	31.3	64.5	43.7
DEAN	55.6	33.8	71.2	49.3
YOLO V8n	53.4	34.9	73.9	52.6
YOLO V8x	55.2	35.6	78.4	55.1
SuperYOLO	59.3	36.4	74.1	53.2
MFE-Framework	49.9	32.5	79.3	58.4
LSK-Net	62.9	47.9	79.6	59.9
TMSI-Net	67.2	46.8	79.5	59.7
DTMSI-Net	**72.3**	**53.1**	**82.3**	**62.7**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.



Fig. 7. Detection results of TMSI-Net and DTMSI-Net in the same image. (a) Comparison of results for the same "Car" target under two detection methods. (b) Comparison of results between two detection methods for the same "Camping car" and "Boat" targets. (c) Comparison of results for the same "Pickup" target under two detection methods. (d) Comparison of results between two detection methods for "Car" targets obstructed by tree branches.

In order to provide a more detailed analysis of the role of the template matching method based on simulation images in remote sensing small target detection tasks and to demonstrate that dynamically optimized simulation images can better capture the geometric and spatial invariant features of targets, we use Faster-RCNN and Modified Darknet-53 as the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.

TABLE V
RESULTS (mAP@0.5/%) OF ABLATION STUDIES

Models	VEDAI	VEDAI-Cloud
Faster-RCNN	41.7	26.6
Faster-RCNN+TMSI	46.0	38.5
Faster-RCNN+DTMSI	58.1	57.6
Modified Darknet-53	60.0	59.9
TMSI-Net (Modified Darknet-53 +TMSI)	67.4	67.2
DTMSI-Net (Modified Darknet-53 +DTMSI)	**76.5**	**72.3**

backbone networks and add TMSI and DTMSI methods to form new detection models. Ablation studies were conducted on the VEDAI and VEDAI-Cloud datasets. The changes in losses and experimental results are shown in Fig. 5 and Table V, respectively. From Fig. 5, it can be seen that whether trained in the VEDAI or VEDAI-Cloud dataset, the model incorporating the DTMSI algorithm has a significantly faster initial convergence speed than other models. Among them, DTMSI-Net converges the fastest and has the least oscillation during the training process.

Comparing the experimental results of the same model on two datasets, it can still be observed that the detection effectiveness of the model decreases after introducing cloud and fog as interference in the dataset. From the comparison of the experimental results in Table V, it is evident that the detection accuracy of the model is higher when the backbone network is selected as Modified Darknet-53 compared to Faster RCNN as the backbone network. This holds for detection results under TMSI, DTMSI, or pure backbone network conditions on the same dataset. This is mainly because key features, such as target position and shape, which play a pivotal role in remote sensing small target detection tasks, are mostly distributed in the shallow layer of the network. The modified Darknet-53 is an optimized model for remote sensing small target detection tasks. From the results of the ablation studies, it can also be seen that the detection accuracy of the two backbone networks gradually improves after adding a template matching method based on simulation images and a dynamic template matching method based on simulation images. This suggests that the template matching method based on simulation images can effectively improve the accuracy of the model in remote sensing small target detection tasks. After tailoring the angle and scale of the simulation images to specific targets as a knowledge template and introducing the dynamic template matching method based on the simulation images into the model, the corresponding remote sensing small target detection task achieves optimal results, regardless of whether there is cloud or fog interference in the dataset.

C. Qualitative Experimental Results

For the same target in different directions in an input image, there are corresponding templates in DTMSI-Net to match it, as shown in Fig. 6(a). The key parameters α and θ for template adjustment are generated through the MLP and T-net modules, and the original template is adjusted to obtain a dynamic template corresponding to the target direction and size. However, the template used in the TMSI Net model is a finite direction template, as shown in Fig. 6(b), which has poor capturing ability for the geometric features and spatial structure of small targets. The comparison in Fig. 6 demonstrates that DTMSI-Net, with its dynamic template adjustment module, exhibits the improved ability to capture the geometric features and spatial structure of targets during the simulation image matching stage, leading to more precise detection results.

The partial detection results of TMSI-Net and DTMSI-Net models are shown in Fig. 7(a)–(d). From the qualitative comparison of detection results, it can be observed that DTMSI-Net with a dynamic template library is less prone to missed or false detections in the detection process of remote sensing small targets and exhibits higher confidence in target detection. In Fig. 7(a), the confidence level of DTMSI-Net for the car is 87.13%, much higher than the 50.63% of the TMSI-Net model. For the analysis of the car, camping car, and boat targets in Fig. 7(b), DTMSI-Net not only exhibited higher confidence in detecting car targets compared to TMSI-Net but also accurately detected camping cars and boats with higher confidence, while the TMSI-Net model completely lost the target. The pickup target in the lower right corner of Fig. 7(c) is severely affected by background interference, making it difficult to distinguish the target edge from the background. Similarly, in Fig. 7(d), there is a tree branch blocking the car target in the lower right corner, which disrupts the structural contour features of the target itself. These noises seriously interfere with the TMSI-Net's detection of targets, resulting in missed detections. For DTMSI-Net, the limited contour of the target can be supplemented and captured by optimizing the direction and size of the simulated image, as shown in Fig. 6(a), ultimately completing the detection task of noisy targets.

V. CONCLUSION

The task of small target detection in remote sensing images is to analyze and understand the basic problems faced by remote sensing images. This problem frequently occurs in many civilian and military scenarios. Due to the poor saliency of target features, special angles, and susceptibility to cloud cover in remote sensing images, this problem has always been difficult to solve. In this article, a dynamic template matching method based on simulation images as knowledge templates is proposed for remote sensing object detection tasks. The target simulation image generated by the simulation framework has clear mathematical and physical models, which can be explained by physical properties. When simulation images are added as templates to neural networks, it is equivalent to prior knowledge or common sense, which can guide the learning and reasoning of the network and improve its accuracy. By dynamically adapting and optimizing the simulation images used as knowledge templates in the model, the DTMSI-Net with a dynamic template library is proposed, and the optimal results are achieved on all VEDAI, VEDAI-Cloud, and NWPU datasets. The results of ablation experiments and qualitative experiments also indicate that the dynamic template matching

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.

method based on simulation images can not only enhance the invariant structural features of remote sensing targets with poor enhancement feature significance and special angles but also, to some extent, supplement missing key geometric and spatial structural features for occluded remote sensing targets. In future research, we will pay more attention to how to combine knowledge of commonsense with neural networks in an interpretable way, which can not only use the powerful learning and processing capabilities of neural networks but also make the network decisions easier to explain.

REFERENCES

[1] D. Li, M. Wang, Z. Dong, X. Shen, and L. Shi, "Earth observation brain (EOB): An intelligent Earth observation system," *Geo-Spatial Inf. Sci.*, vol. 20, no. 2, pp. 134–140, Apr. 2017.

[2] J. Yang, Y. Yang, J. Wen, Y. Li, and S. Ercisli, "Remote sensing image information quality evaluation via node entropy for efficient classification," *Remote Sens.*, vol. 14, no. 17, p. 4400, Sep. 2022.

[3] D. Kopeć, A. Zakrzewska, A. Halladin-Dabrowska, J. Wylazłowska, and Ł. Sławik, "The essence of acquisition time of airborne hyperspectral and on-ground reference data for classification of highly invasive annual vine *Echinocystis Lobata* (Michx.) Torr. & A. Gray," *GISci. Remote Sens.*, vol. 60, no. 1, Dec. 2023, Art. no. 2204682.

[4] H. Schilling, D. Bulatov, R. Niessner, W. Middelmann, and U. Soergel, "Detection of vehicles in multisensor data via multibranch convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4299–4316, Nov. 2018.

[5] Y. Zhang, K. Fu, H. Sun, X. Sun, X. Zheng, and H. Wang, "A multi-model ensemble method based on convolutional neural networks for aircraft detection in large remote sensing images," *Remote Sens. Lett.*, vol. 9, no. 1, pp. 11–20, Jan. 2018.

[6] X. Meng, H. Shen, Q. Yuan, H. Li, L. Zhang, and W. Sun, "Pansharpening for cloud-contaminated very high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2840–2854, May 2019.

[7] W. Sun, L. Tian, Y. Xu, B. Du, and Q. Du, "A randomized subspace learning based anomaly detector for hyperspectral imagery," *Remote Sens.*, vol. 10, no. 3, p. 417, Mar. 2018.

[8] R. Yang, C. Zheng, L. Wang, Y. Zhao, Z. Fu, and Q. Dai, "MAE-BG: Dual-stream boundary optimization for remote sensing image semantic segmentation," *Geocarto Int.*, vol. 38, no. 1, Dec. 2023, Art. no. 2190622.

[9] M. Qiu, L. Huang, and B.-H. Tang, "Bridge detection method for HSRRSIs based on YOLOv5 with a decoupled head," *Int. J. Digit. Earth*, vol. 16, no. 1, pp. 113–129, Oct. 2023.

[10] W. Guo, W. Yang, H. Zhang, and G. Hua, "Geospatial object detection in high resolution satellite images based on multi-scale convolutional neural network," *Remote Sens.*, vol. 10, no. 1, p. 131, Jan. 2018.

[11] X. X. Zhu et al., "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.

[12] Z. Shao, Y. Pan, C. Diao, and J. Cai, "Cloud detection in remote sensing images based on multiscale features-convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 4062–4076, Jun. 2019.

[13] L. Courtrai, M.-T. Pham, and S. Lefèvre, "Small object detection in remote sensing images based on super-resolution with auxiliary generative adversarial networks," *Remote Sens.*, vol. 12, no. 19, p. 3152, Sep. 2020.

[14] F. Zhou, Y. Wu, Y. Dai, and K. Ni, "Robust infrared small target detection via jointly sparse constraint of $l_{1/2}$-metric and dual-graph regularization," *Remote Sens.*, vol. 12, no. 12, p. 1963, Jun. 2020.

[15] X. Pan et al., "Dynamic refinement network for oriented and densely packed object detection," 2020, *arXiv:2005.09973*.

[16] F. Huang, S. Chen, Q. Wang, Y. Chen, and D. Zhang, "Using deep learning in an embedded system for real-time target detection based on images from an unmanned aerial vehicle: Vehicle detection as a case study," *Int. J. Digit. Earth*, vol. 16, no. 1, pp. 910–936, Oct. 2023.

[17] B. Liu, A. Yu, X. Zuo, R. Wang, C. Qiu, and X. Yu, "Deep hierarchical transformer for change detection in high-resolution remote sensing images," *Eur. J. Remote Sens.*, vol. 56, no. 1, Dec. 2023, Art. no. 2196641.

[18] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2005, pp. 886–893.

[19] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.

[20] T. Ojala, M. Pietikäinen, and D. Harwood, "A comparative study of texture measures with classification based on featured distributions," *Pattern Recognit.*, vol. 29, no. 1, pp. 51–59, Jan. 1996.

[21] P. Dollár, Z. Tu, P. Perona, and S. Belongie, "Integral channel features," in *Proc. Brit. Mach. Vis. Conf. (BMVC)*, London, U.K., Sep. 2009, doi: 10.5244/C.23.91.

[22] H. M. Wallach, "Topic modeling: Beyond bag-of-words," in *Proc. 23rd Int. Conf. Mach. Learn. (ICML)*, 2006, pp. 977–984.

[23] T. Tang, S. Zhou, Z. Deng, H. Zou, and L. Lei, "Vehicle detection in aerial images based on region convolutional neural networks and hard negative example mining," *Sensors*, vol. 17, no. 2, p. 336, Feb. 2017.

[24] Y. Chen, H. Xu, X. Zhang, P. Gao, Z. Xu, and X. Huang, "An object detection method for bayberry trees based on an improved YOLO algorithm," *Int. J. Digit. Earth*, vol. 16, no. 1, pp. 781–805, Oct. 2023.

[25] Y. Liu, R. Zhang, R. Deng, and J. Zhao, "Ship detection and classification based on cascaded detection of hull and wake from optical satellite remote sensing imagery," *GISci. Remote Sens.*, vol. 60, no. 1, Dec. 2023, Art. no. 2196159.

[26] Y. Zhang, W. B. Rossow, A. A. Lacis, V. Oinas, and M. I. Mishchenko, "Calculation of radiative fluxes from the surface to top of atmosphere based on ISCCP and other global data sets: Refinements of the radiative transfer model and the input data," *J. Geophys. Res., Atmos.*, vol. 109, no. D19, Oct. 2004, doi: 10.1029/2003JD004457.

[27] H. Luo, C. Chen, L. Fang, K. Khoshelham, and G. Shen, "MS-RRFSegNet: Multiscale regional relation feature segmentation network for semantic segmentation of urban scene point clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8301–8315, Dec. 2020.

[28] H. Lv, Y. Wang, and Y. Shen, "An empirical and radiative transfer model based algorithm to remove thin clouds in visible bands," *Remote Sens. Environ.*, vol. 179, pp. 183–195, Jun. 2016.

[29] Q. Li, W. Lu, and J. Yang, "A hybrid thresholding algorithm for cloud detection on ground-based color images," *J. Atmos. Ocean. Technol.*, vol. 28, no. 10, pp. 1286–1296, Oct. 2011.

[30] Z. Zhu, X. Sun, W. Diao, K. Chen, G. Xu, and K. Fu, "Invariant structure representation for remote sensing object detection based on graph modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022, Art. no. 5625217.

[31] Q. Xiao, B. Liu, Z. Li, W. Ni, Z. Yang, and L. Li, "Progressive data augmentation method for remote sensing ship image classification based on imaging simulation system and neural style transfer," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9176–9186, 2021.

[32] Y. Cao, Z. Yang, H. Wang, X. Peng, C. Gao, and Y. Li, "Template matching based on geometric invariance in deep neural network," *IEEE Access*, vol. 7, pp. 82174–82182, 2019.

[33] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[34] W. Liu et al., "SSD: Single shot MultiBox detector," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 21–37.

[35] Y. Cao, Z. Yang, and C. Gao, "A new method of different neural network depth and feature map size on remote sensing small target detection," *Inteligencia Artif.*, vol. 24, no. 68, pp. 21–32, Sep. 2021.

[36] A.-B. Salberg, "Detection of seals in remote sensing images using features extracted from deep convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 1893–1896.

[37] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[38] Y. Li, Q. Hou, Z. Zheng, M.-M. Cheng, J. Yang, and X. Li, "Large selective kernel network for remote sensing object detection," 2023, *arXiv:2303.09030*.

[39] M.-T. Pham, L. Courtrai, C. Friguet, S. Lefèvre, and A. Baussard, "YOLO-Fine: One-stage detector of small objects under various backgrounds in remote sensing images," *Remote Sens.*, vol. 12, no. 15, p. 2501, Aug. 2020.

[40] Y. Li et al., "A framework of maximum feature exploration oriented remote sensing object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.

[41] J. Qu, Z. Tang, L. Zhang, Y. Zhang, and Z. Zhang, "Remote sensing small object detection network based on attention mechanism and multi-scale feature fusion," *Remote Sens.*, vol. 15, no. 11, p. 2728, May 2023.
[42] C. Zhang, B. Xiong, X. Li, and G. Kuang, "TCD: Task-collaborated detector for oriented objects in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023, Art. no. 4700714.
[43] S. Maweheb, S. Malek, and G. Faouzi, "Geometric invariance in digital imaging for the preservation of cultural heritage in Tunisia," *Digit. Appl. Archaeol. Cultural Heritage*, vol. 3, no. 4, pp. 99–107, 2016.
[44] S. Urooj and S. P. Singh, "Geometric invariant feature extraction of medical images using Hu's invariants," in *Proc. 3rd Int. Conf. Comput. Sustain. Global Develop. (INDIACom)*, Mar. 2016, pp. 1560–1562.
[45] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.
[46] J. Han, J. Ding, N. Xue, and G.-S. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2785–2794.
[47] G. Zhang, S. Lu, and W. Zhang, "CAD-Net: A context-aware detection network for objects in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10015–10024, Dec. 2019.
[48] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2021.
[49] Z. Zhu, X. Sun, W. Diao, K. Chen, G. Xu, and K. Fu, "AOPDet: Automatic organized points detector for precisely localizing objects in aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, Art. no. 5606816.
[50] Y. Tian, X. Li, K. Wang, and F.-Y. Wang, "Training and testing object detectors with virtual images," *IEEE/CAA J. Autom. Sinica*, vol. 5, no. 2, pp. 539–546, Mar. 2018.
[51] W. Zhang et al., "Physics guided remote sensing image synthesis network for ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023, Art. no. 4700814.
[52] W. Dai, Q. Yang, G.-R. Xue, and Y. Yu, "Boosting for transfer learning," in *Proc. 24th Int. Conf. Mach. Learn.*, Jun. 2007, pp. 193–200.
[53] S. Razakarivony and F. Jurie, "Vehicle detection in aerial imagery: A small target detection benchmark," *J. Vis. Commun. Image Represent.*, vol. 34, pp. 187–203, Jan. 2016.
[54] A. L. Ratan and W. E. L. Grimson, "Training templates for scene classification using a few examples," in *Proc. IEEE Workshop Content-Based Access Image Video Libraries*, Jun. 1997, pp. 90–97.
[55] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.
[56] M. Ju, J. Luo, P. Zhang, M. He, and H. Luo, "A simple and efficient network for small target detection," *IEEE Access*, vol. 7, pp. 85771–85781, 2019.
[57] S. Wang, Y. Liu, Z. Lin, and Z. Zhang, "Dual efficient self-attention network for multi-target detection in aerial imagery," *Proc. SPIE*, vol. 11427, Jan. 2020, Art. no. 114270D.
[58] G. Jocher, A. Chaurasia, and J. Qiu. (2023). *YOLO by Ultralytics (Version 8.0.0) [Computer software]*. [Online]. Available: https://github.com/ultralytics/ultralytics
[59] J. Zhang, J. Lei, W. Xie, Z. Fang, Y. Li, and Q. Du, "SuperYOLO: Super resolution assisted object detection in multimodal remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023, Art. no. 5605415.



Yaming Cao received the B.S. degree in automation from the Beijing University of Chemical Technology, Beijing, China, in 2016, and the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing.

He is currently a Lecturer with the School of Computer Science and Technology, North University of China, Taiyuan, China. His research interests include deep learning and interpretability of neural networks.



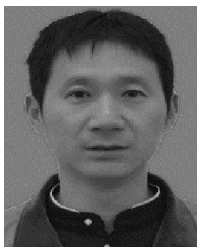
Lei Guo received the B.E. degree in automation, the M.E. degree in control engineering, and the Ph.D. degree in electronic science and technology from the Taiyuan University of Technology, Taiyuan, China, in 2013, 2016, and 2021, respectively.

He is currently a Lecturer with the School of Computer Science and Technology, North University of China, Taiyuan. His main research interests focus on computer vision and incremental learning.



Fengguang Xiong was born in China, in 1979. He received the master's degree in computer application technology from the North University of China, Taiyuan, China, in 2005, and the Ph.D. degree from the Institute of Information Engineering, North University of China, in 2018.

He is currently majoring in computer vision, simulation, and visualization.



Liqun Kuang was born in China, in 1976. He received the master's degree in computer application technology from the North University of China, Taiyuan, China, in 2003, and the Ph.D. degree from the Institute of Information Engineering, North University of China, in 2019.

He is currently majoring in computer vision and pattern recognition.



Xie Han was born in China, in 1964. She received the master's degree in computer science and technology from the North China Institute of Technology (NCIT), Beijing, China, and the Ph.D. degree from the Institute of Information Engineering, University of Science and Technology, Beijing, China, in 2002.

She is currently majoring in computer vision, simulation, and visualization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:16:54 UTC from IEEE Xplore. Restrictions apply.